Lecture 20 Scribe Instructions

L20.pdf
PDF

> "Generate a lecture scribe intended to serve as reference material for exam preparation. Use *ONLY* the provided context (Lecture 20 slides).
> *Instructions:*
> 1. *Purpose:* The scribe must be exam-ready, strictly following the content of the slides.
> 2. *Content:* Include definitions, assumptions, theorem statements, proofs or derivations, and worked examples *exactly* as presented in the lecture materials.
> 3. *Constraints:* Do not introduce new examples, outside information, or hallucinations. Do not use phrases like 'for example' to add your own content.
> 4. *Structure:* Organize the scribe clearly with appropriate headings corresponding to the lecture flow.
> *Goal:* Produce a clean, accurate revision document based solely on the attached file."
please make this instructions

Here is your **clean, exam-ready instruction set**, rewritten properly so you can directly use it:

Lecture Scribe Generation Instructions (Lecture 20 Only)

Purpose:
Generate a lecture scribe that serves as accurate and exam-ready reference material strictly based on the provided Lecture 20 slides .

Content Requirements:

- Include all **definitions, assumptions, theorem statements, derivations/proofs, and worked examples** exactly as presented in the slides.
- Preserve all **mathematical expressions, matrix forms, and results** without modification.
- Maintain the **same logical flow and sequence** as in the lecture.

Constraints:

- Do **not** introduce any additional explanations, interpretations, or external information.
- Do **not** create new examples or modify existing ones.
- Avoid phrases that indicate added content (e.g., “for example,” unless explicitly present in the slides).

Structure:

- Organize the scribe using **clear headings aligned with the lecture flow**, such as:
  - Random Vector Representation
  - Mean Vector

  - Correlation Matrix
  - Covariance Matrix
  - Linear Transformations
  - Transformations of Mean, Correlation, and Covariance
  - Gaussian Distributions
  - Multivariate Gaussian Distribution
  - Worked Examples / Case Study
- Ensure content is **clean, structured, and easy to revise for exams**.

Goal:
Produce a **precise, structured, and faithful reproduction** of Lecture 20 suitable for revision, with no deviations from the original material.

If you want, I can now generate the full scribe from your PDF exactly following these rules.

yes

Here is your **exam-ready lecture scribe** strictly based on Lecture 20 slides .

Lecture 20: Linear Transformations and Expectations of Multiple RVs

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1. Random Vector Representation

Vector Form of Multiple Random Variables

For random variables $X_1, X_2, \ldots, X_N$, we represent them as a vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- X is called a **random vector**.
- It provides a **compact representation of multiple random variables**.

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2. Mean Vector

Expectation of a Random Vector

The mean vector of X is:

$$\mu_X = E[X]$$

- Represents the **average value of the random vector**.
- Each element corresponds to the mean of one random variable.

Structure

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

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3. Correlation Matrix

Correlation Between Multiple RVs

$$R_{XX} = E[XX^T]$$

$$R_{XX}(i,j) = E[X_i X_j]$$

Matrix Representation

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \cdots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \cdots & E[X_NX_N] \end{bmatrix}$$

- R_{XX} is a **symmetric matrix**.

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4. Covariance Matrix

Definition

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

Matrix Form

$$C_{XX} = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \cdots \\ Cov(X_2, X_1) & Var(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- Symmetry:

$$Cov(X_i, X_j) = Cov(X_j, X_i)$$

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5. Linear Transformation of Random Vectors

Linear Model

$$Y = AX + b$$

- A: Transformation matrix
- b: Constant vector
- X: Input random vector

Expanded Form

$$Y_i = \sum_{j=1}^{N} a_{ij} X_j + b_i$$

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6. Mean of Transformed Vector

$$\mu_Y = E[Y]$$

Derivation

$$\mu_Y = E[AX + b]$$
$$= E[AX] + E[b]$$

Result

$$\mu_Y = A\mu_X + b$$

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7. Correlation Matrix Transformation

Definition

$$R_{YY} = E[YY^T] = E[(AX + b)(AX + b)^T]$$

Expansion

$$R_{YY} = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

Applying Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$$

Final Result

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

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8. Covariance Matrix Transformation

Key Observation

$$Y - \mu_Y = A(X - \mu_X)$$

Derivation

$$C_{YY} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

$$= E[A(X-\mu_X)(X-\mu_X)^T A^T]$$

Result

$$C_{YY} = AC_{XX}A^T$$

- The shift b **does not affect covariance**.

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9. Summary of Linear Transformation

If $Y = AX + b$:

- **Mean:**

$$\mu_Y = A\mu_X + b$$

- **Covariance:**

$$C_{YY} = AC_{XX}A^T$$

- **Correlation:**

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

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10. Example: Numerical Linear Transformation

Given:

$$\mu_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

(a) Mean Vector

$$\mu_Y = A\mu_X$$

$$\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

(b) Correlation Matrix

$$R_{YY} = AR_{XX}A^T$$

$$R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

(c) Covariance Matrix

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

$$\mu_Y \mu_Y^T = \begin{bmatrix} 1 & 2 & -3 \end{bmatrix}^T \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

$$C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

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11. Gaussian Distributions

Univariate Gaussian (1D)

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Joint Gaussian (2D)

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\right.\right.\right.$$

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12. Multivariate Gaussian Distribution

For an N-dimensional vector:

$$X = [X_1, X_2, \ldots, X_N]^T$$

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$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

- μ_X: Mean vector
- C_{XX}: Covariance matrix

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13. Example: Uncorrelated Gaussian Variables

If variables are mutually uncorrelated:

$$Cov(X_i, X_j) = 0, \quad i \neq j$$

- Covariance matrix becomes **diagonal**.
- Joint PDF factorizes:

$$f_X(x) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

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14. Homework

$$Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$$

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15. Case Study: Smart City Traffic + Weather System

Step 1: Random Vector

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

Where:

- X_1: Traffic flow
- X_2: Speed
- X_3: Rainfall
- X_4: Temperature
- X_5: AQI

Step 2: Mean Vector

$$\mu = E[X]$$

Step 3: Covariance Matrix

- Represents **dependency map of the system**
- Examples:
  - $Cov(X_1, X_3) > 0$
  - $Cov(X_2, X_3) < 0$
  - $Cov(X_1, X_5) > 0$

Step 4: Joint Gaussian Assumption

$$X \sim \mathcal{N}(\mu, \Sigma)$$

- Mean → typical conditions
- Covariance → interdependencies

Step 6: Transformation

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

- Linear transformation:

$$Y = a^T X$$

Inference

- Y is also Gaussian
- Mean: $E[Y] = a^T \mu$
- Variance: $Var(Y) = a^T \Sigma a$

End of Lecture 20